

TIME, SPEED & DISTANCE

Complete Study Guide, Concepts, 12 Solved Examples & 25 Practice Questions with Answer Key

1. Fundamental Concepts & Theory

Time, Speed, and Distance form one of the most critical pillars of quantitative aptitude. Understanding the core relationship between these three parameters is essential for solving problems ranging from simple journeys to complex relative motion scenarios.

Core Relationships

The fundamental equation connecting speed, distance, and time is:

$$\begin{aligned} \text{Distance} &= \text{Speed} \times \text{Time} \\ \text{Speed} &= \text{Distance} / \text{Time} \quad | \quad \text{Time} = \text{Distance} / \text{Speed} \end{aligned}$$

Units and Conversions

Standard International (SI) units are meters (m) for distance, seconds (s) for time, and meters per second (m/s) for speed. However, in practical applications, kilometers per hour (km/h) is frequently used.

To convert **km/h** to **m/s**, multiply by $5/18$.
To convert **m/s** to **km/h**, multiply by $18/5$.

Proportionality Rules

- **When Distance is Constant:** Speed is inversely proportional to time ($\text{Speed} \propto 1 / \text{Time}$). If speed increases in a ratio of $a:b$, time taken decreases in the ratio $b:a$.
- **When Time is Constant:** Distance is directly proportional to speed ($\text{Distance} \propto \text{Speed}$).
- **When Speed is Constant:** Distance is directly proportional to time ($\text{Distance} \propto \text{Time}$).

Average Speed Concepts

Average speed is defined as the total distance traveled divided by the total time taken:

$$\text{Average Speed} = \text{Total Distance Covered} / \text{Total Time Taken}$$

Special Case (Equal Distances): If a body covers a certain distance at a speed of x km/h and an equal distance at a speed of y km/h, the average speed for the entire journey is given by:

$$\text{Average Speed} = (2xy) / (x + y)$$

Relative Speed

Relative speed is the speed of one moving object with respect to another.

- **Moving in Opposite Directions:** If two bodies move towards each other or in opposite directions with speeds u and v , their relative speed is $u + v$.
- **Moving in the Same Direction:** If two bodies move in the same direction with speeds u and v (where $u > v$), their relative speed is $u - v$.

Trains, Boats & Streams

- **Train Crossing a Stationary Object:** Time = $(\text{Length of train} + \text{Length of object}) / \text{Speed}$ (Length of point object like a pole or man is zero).
- **Boats and Streams:** Let downstream speed (with the current) be $v_d = u + w$ and upstream speed (against the current) be $v_u = u - w$, where u is speed of boat in still water and w is speed of stream. Thus, $u = (v_d + v_u) / 2$ and $w = (v_d - v_u) / 2$.

2. Solved Examples (12 Detailed Problems)

Example 1: Basic Conversion and Distance Calculation

A car is moving at a speed of 72 km/h. How much distance will it cover in 15 minutes?

Solution:

Speed of car = 72 km/h.

Convert speed to m/s or time to hours. Let's convert time to hours: Time $t = 15 / 60 = 1 / 4$ hours.

Distance = $\text{Speed} \times \text{Time} = 72 \times (1 / 4) = 18$ km.

Answer: 18 km

Example 2: Speed Conversion

Express a speed of 25 m/s in kilometers per hour (km/h).

Solution:

To convert m/s to km/h, multiply by $18/5$.

Speed = $25 \times (18 / 5) = 5 \times 18 = 90$ km/h.

Answer: 90 km/h

Example 3: Average Speed with Equal Distances

A person travels from city A to city B at a speed of 40 km/h and returns from B to A at a speed of 60 km/h. Find the average speed for the whole journey.

Solution:

Since the distance traveled in both directions is equal, we can use the formula:

Average Speed = $(2xy) / (x + y)$ where $x = 40$, $y = 60$.

Average Speed = $(2 \times 40 \times 60) / (40 + 60) = 4800 / 100 = 48$ km/h.

Answer: 48 km/h

Example 4: Unusual Time & Speed Variant

Walking at $\frac{3}{4}$ th of his usual speed, a man reaches his office 20 minutes late. Find his usual time to reach the office.

Solution:

Since distance is constant, Speed is inversely proportional to Time. If speed becomes $\frac{3}{4}$ th, time becomes $\frac{4}{3}$ rd of the usual time.

Let usual time be T . Then $(\frac{4}{3})T - T = 20$ minutes.

$(\frac{1}{3})T = 20 \Rightarrow T = 60$ minutes (1 hour).

Answer: 60 minutes

Example 5: Train Crossing a Pole

A train 150 meters long is running at a speed of 54 km/h. In what time will it pass an electric pole?

Solution:

Speed of train = $54 \times (\frac{5}{18}) = 15$ m/s.

Distance to be covered = Length of train = 150 meters.

Time = $\text{Distance} / \text{Speed} = 150 / 15 = 10$ seconds.

Answer: 10 seconds

Example 6: Train Crossing a Platform

A 250-meter-long train running at 72 km/h crosses a platform in 25 seconds. Find the length of the platform.

Solution:

Speed of train = $72 \times (5/18) = 20$ m/s.

Total distance covered in 25 seconds = **Speed** $\times$ **Time** = $20 \times 25 = 500$ meters.

Total Distance = Length of train + Length of platform.

$500 = 250 + \text{Length of platform} \Rightarrow \text{Length of platform} = 250$ meters.

Answer: 250 meters

Example 7: Relative Speed (Opposite Direction)

Two trains of lengths 120m and 180m are running on parallel tracks in opposite directions with speeds 40 km/h and 50 km/h respectively. In what time will they clear each other?

Solution:

Total distance to be covered = $120 + 180 = 300$ meters.

Since they are moving in opposite directions, relative speed = $40 + 50 = 90$ km/h.

Relative speed in m/s = $90 \times (5/18) = 25$ m/s.

Time taken = $300 / 25 = 12$ seconds.

Answer: 12 seconds

Example 8: Relative Speed (Same Direction)

Two persons A and B start running towards each other from two points 60 km apart at speeds of 4 km/h and 6 km/h respectively. After how many hours will they meet?

Solution:

Since they move towards each other (opposite directions), relative speed = $4 + 6 = 10$ km/h.

Total initial distance = 60 km.

Time = **Distance** / **Relative Speed** = $60 / 10 = 6$ hours.

Answer: 6 hours

Example 9: Boats and Streams

A man can row upstream at 8 km/h and downstream at 12 km/h. Find the man's speed in still water and the speed of the current.

Solution:

Let downstream speed $v_d = 12$ km/h and upstream speed $v_u = 8$ km/h.

Speed in still water (u) = $(v_d + v_u) / 2 = (12 + 8) / 2 = 10$ km/h.

Speed of stream (w) = $(v_d - v_u) / 2 = (12 - 8) / 2 = 2$ km/h.

Answer: Still water speed = 10 km/h, Stream speed = 2 km/h

Example 10: Circular Motion (Running in Same Direction)

Two runners A and B run on a circular track of circumference 400 meters with speeds of 10 m/s and 6 m/s in the same direction. When will they meet for the first time at the starting point?

Solution:

Time taken by A to complete one round = $400 / 10 = 40$ seconds.

Time taken by B to complete one round = $400 / 6 = 200/3$ seconds.

To find when they meet at the starting point, we find the LCM of their individual round times: $LCM(40, 200/3) = LCM(40, 200) / HCF(1, 3) = 200 / 1 = 200$ seconds.

Answer: After 200 seconds

Example 11: Thief and Police Chase

A policeman spots a thief at a distance of 200 meters. The thief starts running at 8 km/h and the policeman chases him at 10 km/h. Find the distance run by the thief before he is caught.

Solution:

Relative speed = $10 - 8 = 2$ km/h = $2 \times (5/18) = 5/9$ m/s.

Time taken to catch the thief = $\text{Distance} / \text{Relative Speed} = 200 / (5/9) = 200 \times (9/5) = 360$ seconds (6 minutes).

Distance run by thief = $\text{Speed of thief} \times \text{Time} = (8 \times 5/18) \text{ m/s} \times 360 \text{ s} = (20/9) \times 360 = 800$ meters.

Answer: 800 meters

Example 12: Journey with Stoppages

A bus travels at an average speed of 50 km/h without stoppages, and at 40 km/h with stoppages. Find the average stoppage time of the bus per hour.

Solution:

Due to stoppages, the bus covers 10 km less every hour (difference in speeds = $50 - 40 = 10$ km).
Time taken to cover 10 km at original speed (50 km/h) = $10 / 50 = 1/5$ hour = 12 minutes.

Answer: 12 minutes per hour

3. Practice Questions (25 MCQs)

Q1. A train travels 300 km at a uniform speed. If the speed had been 10 km/h more, it would have taken 1 hour less for the journey. Find the original speed of the train.

- | | |
|------------|------------|
| A) 50 km/h | B) 60 km/h |
| C) 75 km/h | D) 80 km/h |

Q2. A man covers a certain distance by car at 50 km/h and returns back at 30 km/h. What is his average speed for the round trip?

- | | |
|--------------|------------|
| A) 37.5 km/h | B) 40 km/h |
| C) 42.5 km/h | D) 45 km/h |

Q3. If a person walks at 4 km/h, he misses a train by 10 minutes. If he walks at 5 km/h, he reaches 5 minutes before the arrival of the train. Find the distance he walks.

- | | |
|---------|---------|
| A) 3 km | B) 4 km |
| C) 5 km | D) 6 km |

Q4. A train 120 meters long is running at 60 km/h. In what time will it pass a man running at 6 km/h in the same direction?

- | | |
|---------------|---------------|
| A) 6 seconds | B) 8 seconds |
| C) 10 seconds | D) 12 seconds |

Q5. The speed of a boat in still water is 15 km/h and the rate of current is 3 km/h. Find the distance traveled downstream in 12 minutes.

- | | |
|-----------|-----------|
| A) 2.4 km | B) 3.6 km |
| C) 4.0 km | D) 4.2 km |

Q6. A car completes a journey in 10 hours. If it covers the first half of the distance at 40 km/h and the second half at 60 km/h, find the total distance of the journey.

- | | |
|-----------|-----------|
| A) 400 km | B) 450 km |
| C) 480 km | D) 500 km |

Q7. Two stations P and Q are 400 km apart on a straight line. One train starts from P at 7 a.m. and travels towards Q at 50 km/h. Another train starts from Q at 8 a.m. and travels towards P at 60 km/h. At what time will they meet?

- | | |
|---------------|---------------|
| A) 10:00 a.m. | B) 10:30 a.m. |
| C) 11:00 a.m. | D) 11:30 a.m. |

Q8. A thief steals a car at 2:30 p.m. and drives it at 60 km/h. The theft is discovered at 3:00 p.m. and the owner sets off in another car at 75 km/h. When will the owner catch the thief?

- | | |
|--------------|--------------|
| A) 4:30 p.m. | B) 5:00 p.m. |
| C) 5:30 p.m. | D) 6:00 p.m. |

Q9. A man rows downstream 32 km and upstream 14 km, taking 6 hours each time. What is the speed of the current?

- | | |
|-------------|-------------|
| A) 1.0 km/h | B) 1.5 km/h |
| C) 2.0 km/h | D) 2.5 km/h |

Q10. By walking at $\frac{5}{6}$ th of his usual speed, a person is 10 minutes late. Find his usual time.

- | | |
|---------------|---------------|
| A) 40 minutes | B) 50 minutes |
| C) 60 minutes | D) 75 minutes |

Q11. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. What is the ratio of their speeds?

- | | |
|----------|----------|
| A) 1 : 1 | B) 3 : 2 |
| C) 3 : 4 | D) 4 : 3 |

Q12. A bullet is fired from a gun towards a target. A person hears the sound 5 seconds after firing. If the speed of sound is 330 m/s and speed of bullet is 1320 m/s, find the distance of the target.

- | | |
|-----------|-----------|
| A) 1200 m | B) 1320 m |
| C) 1500 m | D) 1650 m |

Q13. A man covers a distance of 80 km in 10 hours partly on foot at 5 km/h and partly on bicycle at 10 km/h. Find the distance covered on foot.

- | | |
|----------|----------|
| A) 20 km | B) 30 km |
| C) 40 km | D) 50 km |

Q14. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of the first train is:

- | | |
|--------------|--------------|
| A) 70 km/h | B) 75 km/h |
| C) 80.5 km/h | D) 87.5 km/h |

Q15. A boy goes to his school at 3 km/h and returns at a speed of 2 km/h. If he takes 5 hours in all, what is the distance between his village and school?

- | | |
|---------|----------|
| A) 6 km | B) 7 km |
| C) 8 km | D) 10 km |

Q16. A train passes a platform 90 meters long in 30 seconds and a pole in 15 seconds. Find the speed of the train.

- | | |
|--------------|--------------|
| A) 15.8 km/h | B) 21.6 km/h |
| C) 24.5 km/h | D) 28.2 km/h |

Q17. In a 100 m race, A can beat B by 10 m and B can beat C by 5 m. By how many meters can A beat C?

- | | |
|-----------|-----------|
| A) 14.5 m | B) 15.0 m |
| C) 15.5 m | D) 16.0 m |

Q18. A motor boat can travel at 10 km/h in still water. It traveled 91 km downstream and returned back to the starting point, taking a total of 20 hours. Find the speed of the stream.

- | | |
|-----------|-------------|
| A) 2 km/h | B) 2.5 km/h |
| C) 3 km/h | D) 3.5 km/h |

Q19. Two cyclists start simultaneously from the same place in opposite directions at speeds of 12 km/h and 15 km/h respectively. After what time will they be 81 km apart?

- A) 2 hours
- B) 2.5 hours
- C) 3 hours
- D) 3.5 hours

Q20. A person travels 600 km by train at 80 km/h, 800 km by ship at 40 km/h, 500 km by aeroplane at 400 km/h and 100 km by car at 50 km/h. What is the average speed for the entire distance?

- A) 60 km/h
- B) 62.5 km/h
- C) 65 km/h
- D) 68 km/h

Q21. A carriage driving in a fog passed a man who was walking at the rate of 3 km/h in the same direction. He could see the carriage for 4 minutes and it was visible to him up to 100 meters. What was the speed of the carriage?

- A) 4.5 km/h
- B) 5 km/h
- C) 5.5 km/h
- D) 6 km/h

Q22. A dog chases a rabbit. The dog takes 5 leaps for every 6 leaps of the rabbit, but 2 leaps of the dog are equal to 3 leaps of the rabbit. Compare the speeds of the dog and the rabbit.

- A) 5 : 4
- B) 4 : 5
- C) 6 : 5
- D) 5 : 6

Q23. Excluding stoppages, the speed of a bus is 54 km/h and including stoppages, it is 45 km/h. For how many minutes does the bus stop per hour?

- A) 9 min
- B) 10 min
- C) 12 min
- D) 15 min

Q24. A man takes 5 hours 45 minutes in walking to a certain place and riding back. He would have gained 2 hours by riding both ways. The time he would take to walk both ways is:

- A) 7 hours 45 mins
- B) 8 hours
- C) 7 hours 30 mins
- D) 8 hours 45 mins

A) 10:32 a.m.

B) 10:48 a.m.

C) 11:00 a.m.

D) 11:15 a.m.